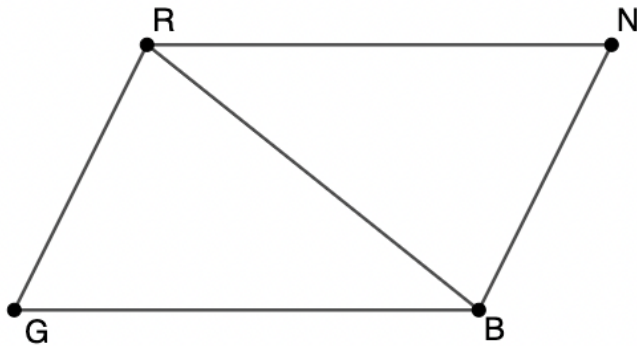


Quadrilateral $RNBG$ is shown with diagonal $\overline{RB}$.



Given: $\overline{RN} \cong \overline{GB}$, $\angle NRB \cong \angle GBR$

Prove: $RNBG$ is a parallelogram.

Statement	Reason
$\overline{RN} \cong \overline{GB}$	Given
1.	2.
$\angle NRB \cong \angle GBR$	Given
$\triangle RGB \cong \triangle RNB$	3.
4.	If two lines are intersected by a transversal and the alternate interior angles are congruent, then the lines are parallel.
5.	6.
$\overline{RG} \parallel \overline{NB}$	If two lines are intersected by a transversal and the alternate interior angles are congruent, then the lines are parallel.
$ABCD$ is a parallelogram	7.

Use the statements and reasons provided to complete the two-column proof.

$\angle BRG \cong \angle RBN$	Corresponding Parts of Congruent Triangles are Congruent
$\angle RGB \cong \angle NBR$	Reflexive Property of Congruence
$\overline{RB} \cong \overline{RB}$	Transitive Property of Congruence
$\overline{RG} \cong \overline{NB}$	$SSS \cong$
$\overline{RN} \parallel \overline{GB}$	$SAS \cong$
$\overline{RG} \parallel \overline{NB}$	A parallelogram has two sets of opposite parallel sides.
	A parallelogram has two sets of opposite congruent sides.

Fill in the blanks with the correct terms to complete the statements.

8. A quadrilateral containing _____ set(s) of _____ sides is a parallelogram.

9. _____ interior angles of a parallelogram are _____.

10. The _____ of a parallelogram bisect each other.

Use the given information to answer questions 1-3.

Quadrilateral $ADHP$ is shown with $HP = 25$, $AD = 8x + 21$, $DB = 20x - 4$ where $x = \frac{1}{2}$.

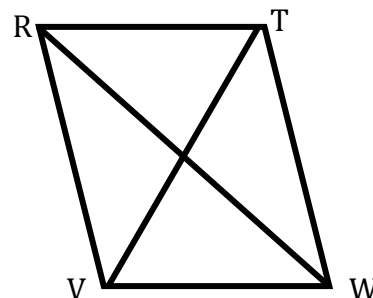


1. I can conclude ...
2. I know this because ...
3. If $ADHP$ is a parallelogram, what is the length of $\overline{PA}$?

Use the given information to answer questions 4-6.

Quadrilateral $RTWV$ is shown with diagonals $\overline{RW}$ and $\overline{TV}$ that intersect at point M where $RM = 4x$, $TM = (3y + 15)$, $WM = 84$, and $VM = 75$.

4. What value of x will make $RTWV$ a parallelogram?

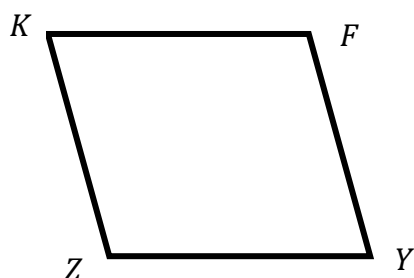


5. What value of y will make $RTWV$ a parallelogram?

6. What is the length of $\overline{VT}$?

Use the given information to answer questions 7-10.

Quadrilateral $KFYZ$ is shown where $m\angle F = (20z - 4)^\circ$, $m\angle Y = 54^\circ$, and $m\angle Z = (8t)^\circ$, where $t = 15\frac{3}{4}$.



Devonte and Briyel are discussing whether $KFYZ$ is a parallelogram. Briyel says it is, and Devonte says it is not.

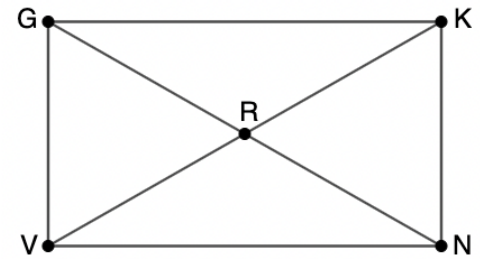
7. Who is correct?

8. Which property of parallelograms was used to determine if $KFYZ$ is a parallelogram?

9. What is the value of z ?

10. What is the $m\angle K$?

Rectangle $GKNV$ is shown with diagonals $\overline{GN}$ and $\overline{KV}$ intersecting at R where $GK = 26x - 13$ and $NV = 14x + 5$.



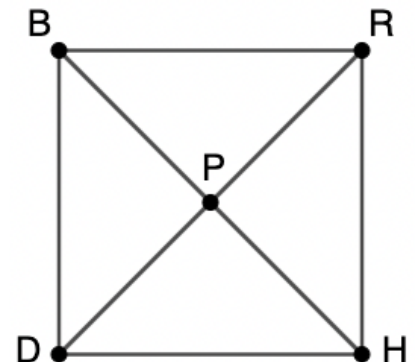
1. Select all true properties of $GKNV$.
 - ☐ Opposite sides are parallel.
 - ☐ Opposite sides are congruent.
 - ☐ Consecutive angles are congruent.
 - ☐ All angles are right angles.
 - ☐ Diagonals are perpendicular.
 - ☐ All sides are congruent.

2. What is the value of x ?

3. What is the length of $\overline{NV}$?

Square $BRHD$ with diagonals $\overline{BH}$ and $\overline{RD}$ intersecting at point P .

4. Write two congruent statements using the diagonals of $BRHD$.

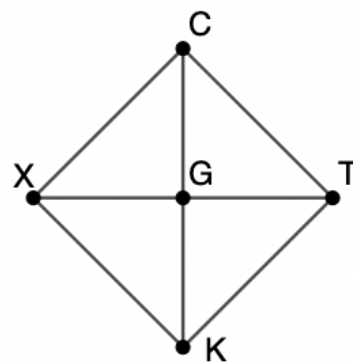


5. Given $BP = 32y - 10$, and $HP = 18y + 32$. What is the value of y ?

6. What is the length of $\overline{RD}$?

Quadrilateral $CTKX$ is shown with diagonals $\overline{CK}$ and $\overline{XT}$ intersecting at point G .

7. What value of g makes $CTKX$ a rhombus, if $CT = 55$, and $TK = 15g + 19$?

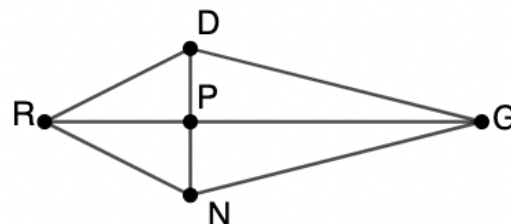


8. Given that $CTKX$ is a rhombus with $CG = 6y - 17.81$ and $KG = 3y + 10.54$. What is the length of $\overline{TX}$?

9. What properties are the same between $GKNV$ and $CTKX$?

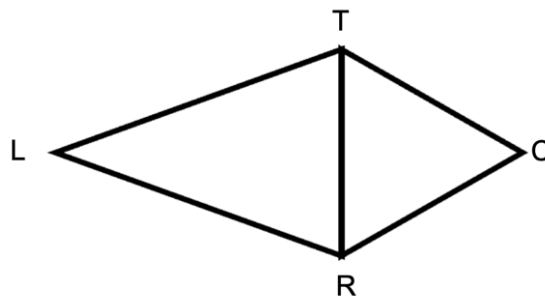
10. What properties are different between $BRHD$ and $CTKX$?

Kite $RDGN$ is shown with diagonals $\overline{RG}$ and $\overline{DN}$ intersecting at point P where $DP = 13$ and $RP = 15$.



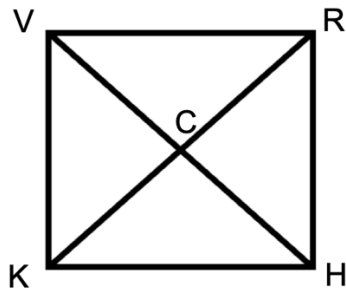
1. What is the length of $\overline{RD}$?
2. Given $PG = 22$. What is the length of $\overline{NG}$?
3. What is the perimeter of $RDGN$?

TCRL is a kite as shown where $m\angle CTR = (7x - 12)^\circ$, $m\angle TCR = (8x + 32)^\circ$, $m\angle LRT = (9y + 40)^\circ$, and $m\angle TLR = (4y)^\circ$.



4. What is the value of x ? Round your answer to the nearest hundredth.
5. What is the value of y ? Round your answer to the nearest hundredth.
6. What is the measure of $\angle CTL$?

Quadrilateral $VRHK$ is shown with diagonals $\overline{VH}$ and $\overline{RK}$ intersecting at point C .



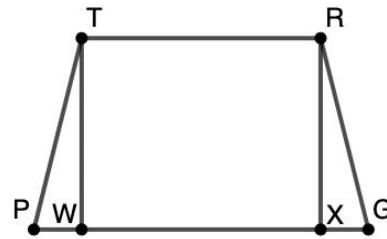
7. What must be true for $VRHK$ to be a square?
8. Given $VRHK$ is a square with $VC = 6z$, and $VR = 12$. What is the value of z ? Leave your answer in radical form.
9. What is the length of $\overline{RK}$, in simplest radical form?
10. What is the length of $\overline{VK}$, in simplest radical form?

Geometry
Unit 8
Lesson 5 Homework

Name _____
Date _____
Period _____

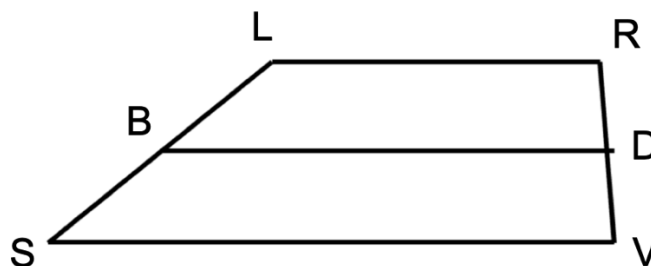
Given: $TRGP$ is an isosceles trapezoid with altitudes $\overline{TW}$ and $\overline{RX}$.

Prove: $\overline{PW} \cong \overline{XG}$



Statement	Reason
$TRGP$ is an isosceles trapezoid	Given
1.	Altitudes of an isosceles trapezoid are congruent.
$\overline{TW} \perp \overline{PG}$ and $\overline{RX} \perp \overline{PG}$	2.
$\angle TWP$ and $\angle RXG$ are right angles.	Perpendicular lines form right angles
3.	All right angles are congruent
4.	5.
6.	AAS Congruence Theorem
$\overline{PW} \cong \overline{XG}$	Corresponding parts of congruent triangles are congruent

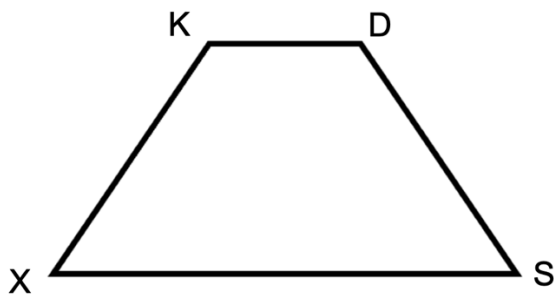
Trapezoid $LRVS$ is shown with the midsegment $\overline{BD}$ where $LB = 5 + 3x$, $SB = 5x - 11$, $RD = 9y + 16$, $VD = 51 + 4y$.



7. What is the value of x and y ?

8. Find the length of $\overline{LB}$ and $\overline{RV}$.

$KDSX$ is an isosceles trapezoid, as shown, where $m\angle K = (35z - 20)^\circ$, $m\angle X = (15z + 8)^\circ$, $m\angle S = (10t + 25)^\circ$.



9. What is the value of z and t ? Round your answer to the nearest hundredth.

10. What is $m\angle S$ and $m\angle D$? Round your answer to the nearest tenth.

Geometry

Unit 8

Lesson 6 Homework

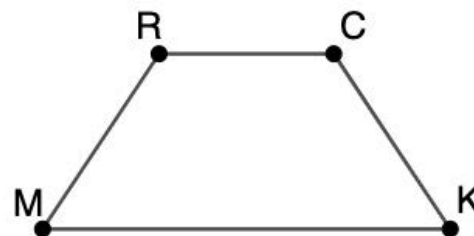
Name _____

Date _____

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Two congruent angles isosceles trapezoid $KMRC$ have measures $(3x + 10)^\circ$ and $(5x - 10)^\circ$.

1. What is the value of x ?



2. What are the measures of all angles of the trapezoid?

For each quadrilateral description, select all shapes that match the description.

3. Quadrilateral $GRMW$ has two pairs of parallel sides and diagonals are congruent.

- ☐ Parallelogram
- ☐ Rectangle
- ☐ Square
- ☐ Rhombus
- ☐ Kite

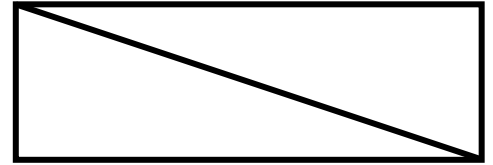
4. Quadrilateral $BNTV$ has diagonals are perpendicular to each other and two pairs of parallel sides.

- ☐ Parallelogram
- ☐ Rectangle
- ☐ Square
- ☐ Rhombus
- ☐ Kite

5. Quadrilateral $KPSX$ has two pairs of congruent sides and diagonals are perpendicular to each other.

- ☐ Parallelogram
- ☐ Rectangle
- ☐ Square
- ☐ Rhombus
- ☐ Kite

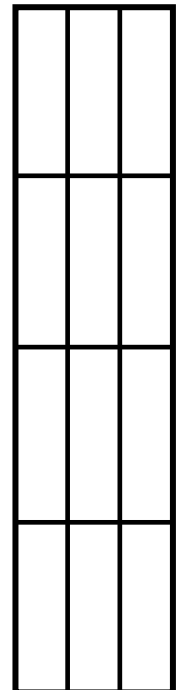
Barrett is looking at a building on the school grounds which is shaped like a rectangle as shown. He knows the building has a height of $25x$ feet and a width of $28x$ feet with a diagonal of 180 feet.



6. What is the value of x ? Round your answer to the nearest hundredth.
7. What is the height of the building? Round your answer to the nearest hundredth.
8. What is the width of the building? Round your answer to the nearest hundredth.

Use the given figure to answer questions 9-10.

A wall has paneling made up of rectangles, with an area of 30 square feet each, as shown. There are four rectangles along the height and three along the top. The edge of each rectangle in the paneling is $(2x + 3)$ feet in width and $(4x + 2)$ feet in height.



9. What is the value of x ?
10. What are the dimensions of the portion of the wall covered the paneling?

Geometry

Unit 8

Lesson 7 Homework

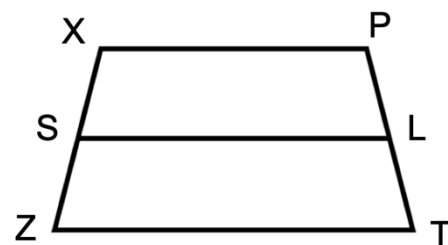
Name_____

Date_____

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$XZTP$ is a trapezoid where $\overline{XP} \parallel \overline{ZT}$. $\overline{SL}$ is a midsegment of $XZTP$ and $XS = 7$, $LT = 6$, $SL = 10$, $TZ = 2x - 4$, and $XP = x - 3$.

1. What is the relationship of $\overline{SL}$ to $\overline{XP}$ and $\overline{ZT}$?



2. What is the value of x ?

3. What is the length of $\overline{XP}$ and $\overline{ZT}$?

Trapezoid $BHLN$ is shown where $\overline{BH} \parallel \overline{NL}$ with midsegment $\overline{DP}$ and $BH = 5x + 12$, $NL = 3x$, and $DP = 5x$.

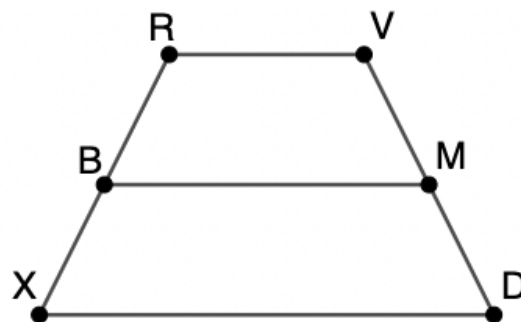
4. Find the value of x .



5. Find the length of $\overline{DP}$.

6. Find the lengths of $\overline{BH}$ and $\overline{NL}$.

Isosceles trapezoid $RVDX$ is shown with midsegment $\overline{BM}$ where $RV = 2x + 1$, $XD = 3x + 2$, $BM = 2x + 4$ and $MD = 3$.



7. What is the value of x ?

8. What is the length of $\overline{BM}$?

9. What is the length of $\overline{XD}$ and $\overline{RV}$?

10. What is the perimeter of $RVDZ$?